A1Q3

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Q3

Q&A:

a) (2 marks) In your own words. Assuming that the Chicago Quantum Ratio (CQR) is meant to be maximized, what two features of the ideal stock portfolio does this imply?

A: The maximized CQR implies the stocks have high expected return as well as low risk (quantified by low standard deviation).

b) (1 mark) What is the size of the population of candiate 60-asset portfolios?

A: There are approximately 1.15 * (10ˆ18) possible combinations.

c) (2 marks) In your own words (30 words or fewer), how does the genetic algorithm being applied to stocks work in this instance. Not every detail is needed.

A: It creates new portfolios by combining an mutating top-scoring ones, then selects the best one using CQNS, repeating this process over generations to improve results.

d) (1 mark) Does a hotter 'temperature' in a simulated annealer imply that the solution is more or less likely to settle into a local minima? No explanation is necessary.

A: Less likely.

e) (1 mark) Is Chicago Quantum Net Score being minimized or maximized by the D-Wave system in this application? No explanation is necessary.

A: Minimized

f) (1 mark) What tendency did the D-Wave system have when finding portfolios that differed from what was desired? No explanation is necessary.

A: It tended to return smaller portfolios than expected.

g) (2 marks) In your own word, in comparative analysis, how well did the D-Wave system do at finding the best possible portfolio?

A: It successfully found efficient and sometimes even ideal portfolio, often outperforming random methods at high-risk levels, but do struggled with the large-sized portfolios.